Mixed spectra of the first memory fields
An upper-charge lift and a positive three-field measure

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Abstract

We close the two-energy cross measure between the first memory tangent and the projected second memory coefficient. The cubic residual has vanishing cross with the creation-force direction, as proved by a complete endpoint certificate. A direct commutator lift handles the upper-charge sector in its cross with the linear rank-filtered field; the same signed-covariance argument extends to the quartic residual. These results yield a positive spectral matrix for three memory fields and the covariance of an explicitly projected quadratic polynomial in memory time.

1 The completed mixed object

We use standard CAR, the full overlapping complex Gaussian p-body ensemble, $h = H_m$, fixed half filling $m = N/2$, normalized maximally mixed trace, and annealed inner product $\langle U, V\rangle = \mathbb{E}\,\mathrm{tr}_m\, U^\dagger V$. Set $\hbar = 1$ and take

$$\lambda_N = p^2/N \longrightarrow \lambda > 0, \qquad q = e^{-4\lambda}, \qquad C_N = \mathbb{E}\,\mathrm{tr}_m\, h^2 \longrightarrow C = J_s^2/\lambda. \tag{1}$$

The fields $Y = Y_-$, R, F, and Q_4 are those of the preceding notes [3]. In particular, with $J_h U = \{h, U\}/2$,

$$L_N = R_N + aJ_hY_N + F_N + o_{L^2}(1), \qquad a = -\frac{(1-q)^2}{1+q^2}. \tag{2}$$

$L_N = mZ_1 - C_{1,m}$ is the conserved-subtracted first memory coefficient. Here L names a field, not the commutator with h. For the projected second coefficient,

$$X_N = (1 - \widehat{\Pi}_N)mZ_2 = Q_{4,N} + 8\lambda_N C_N q_N^2 T_N + o_{L^2}(1), \tag{3}$$

where $\widehat{\Pi}_N$ projects onto $\mathrm{span}\{I, B, O_-, J_hF, E_0, M_0, S_0, Z_0\}$. The main result is the full two-energy statement

$$\boxed{d\nu_{XL} = d\nu_{XY} = 0.} \tag{4}$$

The Y cross was established previously; this note supplies the three missing ingredients for the L cross. Multiplication by any fixed polynomial of the left and right energies preserves these zero cross measures.

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2 Creation force versus the cubic residual

Write $d_i = [H, c_i]$, $A_- = \sum_i d_i^\dagger d_i$, $A_+ = \sum_i d_i d_i^\dagger$, $O_\pm = \mathsf{W}_2 A_\pm$, $B = \mathsf{W}_2 h^2$, and

$$Y_\pm = O_\pm - cB, \qquad c = \frac{2\lambda(1-q)}{1+q}. \tag{5}$$

All W_j in this note are coupling-coordinate Gaussian chaos. The earlier strong output-deficit bridge gives

$$T_N = \frac{Y_- + Y_+}{2} - \rho Y_- + o_{L^2}(1), \qquad \rho = \frac{v}{v + d_v}, \tag{6}$$

with $v = 2\lambda(1-q) + 16\lambda^2 q/(1+q)$ and $d_v = 2\lambda(1+q)$. Thus ν_{TR} reduces to the two oriented force crosses.

For the cubic field put $D_3 = (1+q^2)(1+q+q^2)$. In the lexicographic three-slot mark order $01, 02, 12$, the numerator weights of R are

$$[-q(1+q+q^2),\ D_3,\ -q(1+q+q^2)], \tag{7}$$

and its unmarked numerator is $-2\lambda(1-q)(1+q^3)$. For Y_+ the two-slot mark numerator is $1+q$ and the unmarked numerator is $-2\lambda(1-q)$, with denominator $1+q$.

Take a cubic normal mark (a, b) and a quadratic reverse mark (c, d). For Gaussian pairing π, the established local rule is

$$\alpha = 2 - 4\mathbf{1}_{\text{incident crossing}}, \qquad \delta_{-+} = -4\mathbf{1}_{\pi(a)=c,\,\pi(b)=d}. \tag{8}$$

The quadratic coefficient is $\alpha\beta$ in both orientations; the equal-orientation linear coefficient is instead $\delta_{--} = 4\mathbf{1}_{\pi(a)=d,\pi(b)=c}$. After the Gaussian chaos restrictions, the three- and two-slot blocks share $k = 0, 1, 2$ bridges. Classify the remaining endpoints by their A/B colors and the split between exterior arcs. The number of templates is

$$\sum_{k=0}^{2} \binom{5-2k}{3-k}(6-2k) = 74. \tag{9}$$

Each contains $3!2!/k!$ pairings, for 876 words altogether. Every coefficient of λ and λ^2 vanishes as an integer polynomial in q, for either force orientation. The code also checks all 2628 local two-mark minors per orientation against the determinant implementation.

Exterior spectator chords contribute the same factor on each endpoint orbit. The certificate therefore covers arbitrary fixed two-arc energy degrees, not just bounded word lengths. Positive matrix-measure compactness and moment uniqueness, using the established diagonal kernels, give

$$\boxed{\nu_{R,Y_+} = \nu_{R,Y_-} = \nu_{R,T} = 0.} \tag{10}$$

3 The filtered edge must also reach the upper charge

Write the neutral rank decomposition $h = \sum_a h_a$, with $\mathbb{E}\,\mathrm{tr}_m\, h_a^2 = C_N \omega_a^{(p)}$, and put $u = p - a$. The earlier normalized linear residual satisfies

$$f_N = F_N/\sqrt{V_{F,N}}, \qquad \widehat{f}_N = \sqrt{\lambda_N}\sum_a [R_1(p-a)-1]h_a/\sqrt{C_N}, \tag{11}$$

$$R_1(u) = \frac{u(N-2p+u+1)(N-m+p)}{(m-p+1)p^2}, \quad \|f_N - \widehat{f}_N\|_2 \longrightarrow 0. \tag{12}$$

This uses the exact rank-weight ratio, factorial input tails, and the Poisson deficit limit; it is an operator L^2 replacement. A Wick edge from $\widehat{f}_N$ is $\sqrt{\lambda_N}(\mathsf{K}_{p-1} - \mathsf{K}_p)$ at charge m.

Let U be a $(p-1)$-body ensemble normalized at m, and set

$$d = m - p + 1, \qquad g(n) = \frac{n-p+1}{d}, \qquad \widetilde{U} = g(Q)U. \tag{13}$$

For every allowed charge n, the exact reduced rank factors obey

$$g(n)\frac{\binom{n-a}{p-1-a}}{\binom{m-a}{p-1-a}} = \frac{\binom{n-a}{p-a}}{\binom{m-a}{p-a}}. \tag{14}$$

Together with R_1, this extends the signed covariance to $m \leftrightarrow n$. The new finite tensor check explicitly includes $n = m+1$, which the creation-force vertex requires.

Define bilinear commutator marks for neutral U, V:

$$\mathcal{M}_-(U,V) = -\sum_i [U, c_i^\dagger][V, c_i], \qquad \mathcal{M}_+(U,V) = -\sum_i [U, c_i][V, c_i^\dagger]. \tag{15}$$

For $DV = QV - \sum c_i^\dagger V c_i$ and $\overline{D}V = (N-Q)V - \sum c_i V c_i^\dagger$, direct CAR gives at charge m

$$\mathcal{M}_+(gU,V) = (1+d^{-1})\mathcal{M}_+(U,V) - d^{-1}U\overline{D}V, \tag{16}$$

$$\mathcal{M}_+(V,gU) = (1+d^{-1})\mathcal{M}_+(V,U) - d^{-1}(\overline{D}V)U, \tag{17}$$

$$\mathcal{M}_-(gU,V) = (1-d^{-1})\mathcal{M}_-(U,V) + d^{-1}UDV, \tag{18}$$

$$\mathcal{M}_-(V,gU) = (1-d^{-1})\mathcal{M}_-(V,U) + d^{-1}(DV)U. \tag{19}$$

These formulas preserve operator ordering. For example $[gU, c] = g[U,c] - d^{-1}cU$; the sum in the remaining term reduces to $(\overline{D}U)V - \overline{D}(UV)$, proving the first equation. For a p-body V, $DV = pV$ and $\overline{D}V = pV - \Delta_V$. At half filling Δ_V has bounded fixed moments by the previous input-deficit formula. Hence every correction in (17) is $O_{L^2}(p/N) = O_{L^2}(p^{-1})$, also inside each fixed-degree trace by higher-moment bounds.

4 Why the filtered crosses vanish at all energy orders

It is essential to apply the preceding identities to the actual reverse commutator mark. The general superoperator formula is

$$\overline{D}[f(Q)B]_m = f(m+1)(\overline{D}B)_m + (N-m)[f(m) - f(m+1)]B_m. \tag{20}$$

Replacing a normal body-count mark mechanically by $\overline{D}$ would leave a correction of order one. Equation (17) includes the needed terms directly and has the smaller error just proved.

For normal marks in Q_4, the old lifting identity extends to every fixed product degree k. If B contains one $(p-1)$-body factor and $k-1$ factors of body order p, then

$$[(kp - D)(gB)]_m = (1 - d^{-1})[(kp - 1 - D)B]_m + \frac{(k-1)p}{d} B_m. \tag{21}$$

In particular $k = 4$ treats the quartic vertex. Nonadjacent pair marks can be written as finite sums of total marks on contiguous subproducts, so (21) also treats each of its six pair positions. If the shifted slot lies outside the mark, the scalar lift evaluates to one at the exterior charge. No growing number of factors is used.

The mixed-body chord limit in Ref. [1] applies to index sizes p and $p-1$: their pair-overlap means have the same limit, and triple overlaps vanish. The project's marked determinant calculation, together with absolute exponential overlap bounds, permits fixed mark derivatives of this limit. The half-filled trace bridge transfers the argument to the present state. Consequently each fixed marked graph has the same leading value when one covariance edge is shifted from p to the lifted $p-1$ ensemble. Equations (17) and (21) control the additional vertex terms.

For $B_N = Y_{+,N}$ or $Q_{4,N}$ and fixed r, s, replacing F_N by its normalized proxy in (12) is legitimate because

$$|\langle (f_N - \widehat{f}_N) h^r B_N h^s \rangle| \le \|f_N - \widehat{f}_N\|_2 (\mathbb{E}\,\mathrm{tr}_m\, B_N h^{2r} B_N h^{2s})^{1/2} \longrightarrow 0. \tag{22}$$

The second factor is bounded by the already established diagonal energy moments. Gaussian centering deletes internal pairings of the Y_+ block; W_4 deletes those of the quartic block. Each surviving Wick graph has exactly one signed filtered edge. Its two lifted terms have identical limits and cancel. Thus all fixed mixed moments vanish. The positive spectral matrix, diagonal high moments, and compact candidate support then imply

$$\boxed{\nu_{F,Y_+} = \nu_{F,Q_4} = 0, \qquad \nu_{F,T} = 0.} \tag{23}$$

The last statement uses the already known $\nu_{F,Y_-} = 0$. Equations (10) and (23), the old $\nu_{Q_4,R} = 0$, and the preceding $\nu_{X,Y_-} = 0$ prove (4).

5 A positive three-field matrix and its projected polynomial

Let μ_r be the existing normalized Mehler two-energy measure, with the variance-C energy marginal, and put $e = (E + E')/2$. The complete matrix for (Y, L, X) is

$$d\boldsymbol{\nu} = \begin{pmatrix} W\, d\mu_{q^2} & aeW\, d\mu_{q^2} & 0 \\ aeW\, d\mu_{q^2} & d\nu_L & 0 \\ 0 & 0 & d\nu_X \end{pmatrix}, \tag{24}$$

where

$$d\nu_L = V_F d\mu_q + V_R d\mu_{q^3} + a^2 e^2 W d\mu_{q^2}, \tag{25}$$

$$d\nu_X = V_4 d\mu_{q^4} + V_2 d\mu_{q^2}, \tag{26}$$

$$W = 4J_s^4 \left(\lambda^{-1} + \frac{4q}{1+q}\right), \qquad V_F = 64 J_s^6 q^2, \tag{27}$$

$$V_R = \frac{C^3}{D_3^2}(\lambda A_1 + \lambda^2 A_2), \tag{28}$$

$$A_1 = 4(1-q^2)(1+q^2)(1+q+q^2)^2, \tag{29}$$

$$A_2 = 16q^2(1+q)(1+q^2)(1+q+q^2)(2q^2-q+2). \tag{30}$$

The previous note gives the positive V_4, V_2 explicitly. Relative to the common product energy marginal, write $K_j = d\mu_{q^j}/d(\nu_q \otimes \nu_q)$. Then (24) has the exact positive factorization

$$M\ \mathrm{diag}(WK_2, V_F K_1, V_R K_3, V_4 K_4, V_2 K_2) M^T, \quad M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ ae & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}. \tag{31}$$

The determinant is positive:

$$WK_2(V_F K_1 + V_R K_3)(V_4 K_4 + V_2 K_2) > 0.$$

The five fields Y, T, F, R, Q_4 have diagonal two-point measures; Y and T share the normalized q^2 kernel.

Define the explicitly filtered polynomial $\Phi(s) = Y + isL - s^2 X/2$. Its energy-resolved covariance is

$$\begin{aligned} d\nu_{\Phi(s),\Phi(u)} &= W[1 + iae(u-s) + a^2 e^2 su] d\mu_{q^2} \\ &\quad + su(V_F d\mu_q + V_R d\mu_{q^3}) + \frac{s^2 u^2}{4} d\nu_X. \end{aligned} \tag{32}$$

At ordinary center time T, reflection $(E, E') \mapsto (-E', -E)$ preserves the frequency and reverses e. Thus

$$\langle \Phi(s)^\dagger \Phi(u; T) \rangle = W G_{q^2}(T) + suC_L(T) + \frac{s^2 u^2}{4} C_X(T). \tag{33}$$

Since X has an additional, partly nonconserved projection, Φ is not the raw memory Taylor polynomial. Formula (32) does not assert a finite-memory remainder bound. Such a bound must also restore the removed dynamic descendants.

6 Checks, scope, and the next reconstruction step

Endpoint certificate. Both orientations were computed, rather than extrapolating the new result from the old normal one: 74 templates and 876 words per orientation, 5256 determinant-minor comparisons in total. All integer polynomial arrays vanish. The direct certificate also gives a shorter route to the previously known R–Y_- cross; it is used here as a control, not claimed as a new kernel.

Upper-charge and operator checks. The exact rank-Casimir covariance tensors were evaluated for $(N, p, m) = (4, 2, 2), (5, 2, 3), (6, 2, 3)$, now explicitly including $m + 1$. Their 93076 entries agree after clearing denominators. The previous normal quadratic/cubic controls were retained. Independent integer Hermitian samples test both locations of a lifted factor in each CAR orientation, the reverse superoperator identity, and all four locations in a noncommuting quartic product: 81 exact identities, with 80 nonzero witnesses. Cleared entries remain below 2^{48}. No Gaussian sampling is used to estimate ensemble expectations; the samples only check operator identities.

The first checker required every witness to be nonzero. It stopped on the auxiliary $\overline{D}(gU)$ identity for one traceless one-body sample at $(6, 2, 3)$. Here $\overline{D}U = U - \mathrm{tr}(J)I$ and its cleared identity reduces to $-3\,\mathrm{tr}(J)I$, so zero is correct. The original checker and diagnosis are preserved. The revised checker records this zero and still requires all reverse/normal commutator and quartic witnesses to be nonzero. No algebraic tolerance was relaxed.

Assembly checks. An independent symbolic multiplication verifies the positive factorization, determinant, all nine polynomial coefficients in (s, u), the conjugation symmetry, and the energy-reflection reduction. Existing diagonal-kernel calculations supply the time kernels; no new diagonal numerical scan is needed. Every new scientific output records its inputs, hashes, and environment.

What follows. The missing mixed spectra of the specified three fields are now closed at fixed half filling. The next work must extend the projection and bounds uniformly in a central charge window, restore the removed dynamic descendants when constructing the actual second memory coefficient, and control the two-parameter finite-memory remainder. The resulting kernel, rather than the projected polynomial alone, is the object to connect to the Susskind singlet/meson spatial reconstruction program. Two-point spectral orthogonality does not establish mixed higher-point independence or a bulk mass dictionary.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Sec. 3.3, Eqs. (3.11)–(3.13): oriented words with generic index-set lengths.

[2] J. Tall and S. Tomsovic, *Double-scaled bosonic and fermionic embedded ensembles, complex SYK, and the dual Hilbert space*, arXiv:2604.14522v2. Existing spectral Wick and strength-density framework.

[3] A. Day and Chengheng, local project notes (2026-11-06): *Closing the cubic singlet two-point spectrum*; *Energy-resolved memory background and first fluctuations*; *Two orientations of the fermionic force*. The corresponding source, exact certificates, and PDFs remain in the project.